

UGOLNYE KOPI, Russian Federation

WMO: 255630

Lat: 64.735N

Lon: 177.741E

Elev: 59

StdP: 100.62

Time Zone: 12.00 (E12)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.4	-35.6	-39.8	0.1	-37.3	-38.0	0.1	-35.5	25.0	-8.4	23.1	-7.3	6.3	0	1.015

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.3	20.9	15.1	18.6	13.8	16.7	12.8	15.5	19.6	14.2	17.6	13.2	16.1	5.0	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.6	9.8	17.5	12.5	9.1	16.1	11.5	8.5	14.9	43.8	19.6	40.2	17.7	37.4	16.2	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.6	17.8	14.6	DB	-39.4	24.6	2.1	1.7	-40.9	25.8	-42.2	26.8	-43.4	27.8	-44.9	29.0	(4)
(5)			WB	-39.2	17.7	2.3	1.3	-40.8	18.6	-42.2	19.4	-43.5	20.1	-45.2	21.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-5.8	-21.8	-20.8	-18.0	-11.7	-0.7	7.8	12.4	10.8	5.7	-3.6	-11.6	-18.5
	DBStd	13.99	9.18	9.59	8.02	6.52	4.36	4.16	3.19	3.07	3.65	5.41	7.90	9.71
	HDD10.0	5917	987	862	867	650	333	91	12	27	134	422	648	884
	HDD18.3	8793	1245	1096	1126	900	591	315	187	234	378	681	898	1142
	CDD10.0	167	0	0	0	0	0	26	85	51	6	0	0	0
	CDD18.3	2	0	0	0	0	0	0	2	1	0	0	0	0
	CDH23.3	11	0	0	0	0	0	2	7	1	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.3	6.7	7.1	6.8	5.8	5.5	5.8	5.8	5.4	5.4	6.0	7.6	7.1
Precipitation	PrecAvg	344	42	36	31	20	11	19	37	43	30	25	27	34
	PrecMax	616	116	147	103	83	34	60	104	87	92	80	80	128
	PrecMin	159	2	1	0	1	2	3	5	5	0	1	1	0
	PrecStd	96	26	32	28	18	7	11	24	21	21	17	21	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	0.2	0.7	2.0	11.8	22.9	24.2	22.0	16.2	8.2	3.2	1.1
		MCWB	-0.1	-0.3	-0.1	0.4	7.1	14.5	16.9	15.9	11.9	7.1	2.6	0.3
	2%	DB	-1.0	-1.0	-0.8	0.4	8.1	19.1	21.7	19.0	13.8	7.0	1.9	0.0
		MCWB	-1.7	-1.4	-1.3	-0.5	5.0	12.7	15.9	14.2	10.1	6.3	1.3	-0.5
	5%	DB	-3.9	-2.8	-2.2	-0.6	5.9	16.7	19.4	16.9	11.9	4.8	0.7	-1.2
		MCWB	-4.4	-3.5	-2.8	-1.3	3.4	11.5	14.6	13.2	9.4	3.8	0.2	-1.8
	10%	DB	-7.4	-5.2	-5.3	-2.1	4.0	14.2	17.3	15.1	10.7	3.1	-0.5	-3.2
		MCWB	-8.1	-5.7	-6.0	-2.9	2.3	10.1	13.5	12.1	8.8	2.1	-1.0	-3.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	-0.3	0.0	0.7	7.2	15.0	17.8	16.9	12.8	7.5	2.9	0.6
		MCDB	0.2	-0.1	0.3	1.2	10.6	21.3	23.3	20.8	15.5	8.0	3.3	0.8
	2%	WB	-1.8	-1.5	-1.3	-0.4	5.1	13.2	16.2	14.7	10.8	6.1	1.2	-0.5
		MCDB	-1.2	-1.1	-0.7	0.2	7.8	18.4	20.7	17.9	12.6	6.9	1.5	-0.1
	5%	WB	-4.4	-3.4	-3.0	-1.3	3.5	11.7	15.0	13.6	9.9	3.8	0.2	-1.9
		MCDB	-4.0	-2.8	-2.3	-0.6	5.4	16.2	18.7	16.2	11.4	4.5	0.5	-1.3
	10%	WB	-8.1	-5.7	-6.0	-3.0	2.4	10.3	13.9	12.5	9.0	2.1	-1.0	-3.8
		MCDB	-7.5	-5.1	-5.5	-2.2	3.8	13.9	16.8	14.7	10.5	2.9	-0.5	-3.1
Mean Daily Temperature Range	5% DB	MDBR	6.6	6.9	7.1	7.3	5.9	7.3	6.3	5.9	5.5	4.9	6.0	6.5
		MCDBR	8.3	7.4	6.8	5.8	8.6	11.4	9.6	8.5	7.2	4.6	5.7	6.4
		MCWBR	8.2	7.3	6.6	5.2	5.9	6.6	5.5	5.0	4.8	3.9	5.7	6.3
	5% WB	MCDBR	8.3	7.6	6.7	5.4	8.0	10.7	8.7	7.6	5.7	4.0	6.1	6.7
		MCWBR	8.3	7.6	6.7	5.0	5.7	6.4	5.4	4.9	4.2	3.7	6.1	6.7
Clear-Sky Solar Irradiance	taub	0.169	0.245	0.299	0.351	0.403	0.382	0.447	0.415	0.344	0.287	0.198	0.181	
	taud	1.929	2.133	2.068	1.952	1.973	2.293	2.129	2.196	2.370	2.254	1.948	1.927	
	Ebn at Noon	392	663	763	798	790	830	750	737	728	596	310	115	
	Edh at Noon	24	58	98	140	152	114	131	111	75	52	22	8	
All-Sky Solar Radiation	RadAvg	0.14	0.78	2.49	4.64	5.65	5.64	4.92	3.71	2.27	0.90	0.23	0.04	
	RadStd	0.02	0.12	0.25	0.19	0.43	0.43	0.49	0.37	0.25	0.11	0.03	0.00	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.60	N/A	N/A	+1.55	+0.92	+1.00	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page